

CHAMODA DASANAYAKE

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Chamoda Dasanayake | AI Engineer & Full-Stack Developer

PROFESSIONAL SUMMARY

Final-year Electronics and Computer Science undergraduate with strong foundations in software engineering and AI-driven application development. Experienced in designing and deploying scalable web applications, RESTful APIs, and intelligent systems using Python, FastAPI, React, Node.js, Spring Boot, C#, .NET, and SQL/NoSQL databases. Skilled in frontend-backend integration, machine learning workflows, and modern software architecture. Proven ability to build end-to-end projects. Passionate about developing scalable, high-performance software solutions and advancing in AI/ML engineering and full-stack software development roles.

EDUCATION

BSc (Hons) in Electronics and Computer Science

Jul 2023 - Feb 2027

University of Kelaniya

Current GPA: 3.5 / 4.0

Relevant Coursework: Machine Learning, Advanced Database Systems, Enterprise Software Architecture, Full-stack Development, Software Engineering, Communication Systems, Microcontrollers, Control Systems

Diploma in Information Technology (NVQ Level 5)

Jan 2022 - Jul 2023

Vocational Training Authority of Sri Lanka

Focused Areas: Software Development, Database Systems, Networking, Programming Fundamentals, Cyber Security

G.C.E. Advanced Level (A/L) – Physical Science Stream

Feb 2021

St. Joseph's Balika Maha Vidyalaya, Kegalle

Results: 3 B passes

District Rank: 129

G.C.E. Ordinary Level (O/L)

Aug 2017

Kegalle Balika Vidyalaya

Results: 9A passes

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, C, C#
- **Frontend Development:** React.js, HTML5, CSS3, TypeScript
- **Backend & API Development:** Node.js, Express.js, Spring Boot, FastAPI, Flask, RESTful API Design
- **Databases:** MySQL, MongoDB, Microsoft SQL Server, Oracle (PL/SQL), ChromaDB (Vector Database)
- **AI & ML:** Machine Learning (Scikit-learn), Deep Learning (PyTorch, TensorFlow, Keras), NLP, LLMs, RAG Systems, Embeddings & Vector Search, Prompt Engineering, Recommendation Systems, Feature Engineering, Model Evaluation & Optimization
- **Frameworks & Libraries:** LangChain, LangGraph, CrewAI, LlamaIndex
- **Infrastructure & Deployment:** Docker, AWS EC2, Nginx, PM2, Git, GitHub, CI/CD Pipelines, Vercel, n8n, MLflow, MLOps
- **Security & Authentication:** JWT Authentication, REST API Security Basics
- **Software Engineering & System Design:** System Design Fundamentals (LLD & HLD basics), Microservices Design, Agile Development, CRUD System Design, API Design, Debugging, Unit Testing, Data Structures & Algorithms

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI, Stanford University
Credential ID: THRYI7UH1S6P
- Unsupervised Learning, Recommenders, Reinforcement Learning – DeepLearning.AI
Credential ID: ORVL9CURU1E3
- AI/ML Engineer - Stage 1 - Sri Lanka Institute of Information Technology (SLIIT)

PROJECTS

Multi-Agent Conversational RAG System

Tech Stack: Python, CrewAI, LlamaIndex, ChromaDB, FastAPI, Streamlit, Hugging Face Transformers

- Architected a Multi-Agent RAG pipeline using CrewAI and LlamaIndex with retrieval, fact-checking, and response refinement agents to generate grounded, hallucination-controlled document responses.
- Implemented LLM-based query coreference resolution to improve context-aware retrieval accuracy for sequential conversational Q&A.
- Developed a voice-enabled Streamlit AI dashboard featuring drag-and-drop document ingestion, hallucination audit visualization, and real-time response playback.

Scalable MERN AI Chatbot Platform with LLM Integration

Tech Stack: React, Node.js, Express, MongoDB, LLM APIs, AWS, Nginx, PM2

- Built a production-grade full-stack AI chatbot platform integrating Google Gemini LLM for real-time conversational intelligence with persistent chat memory and context-aware responses.
- Designed a secure and scalable backend architecture using Node.js, Express, MongoDB, JWT authentication, and HTTP-only cookies for session management and protected API communication.
- Deployed and optimized a cloud-ready AI system on AWS EC2 using Nginx and PM2, ensuring high availability, low latency, and support for concurrent real-time user interactions.

Production-Ready NLP Text Summarization System

Tech Stack: Python, FastAPI, Hugging Face Transformers, MLflow, DVC

- Built a production-ready NLP text summarization system using Hugging Face Transformer models to generate abstractive summaries for long-form text and PDF documents with real-time inference capability.
- Developed an end-to-end MLOps pipeline covering data ingestion, preprocessing, model training, evaluation, and deployment using FastAPI for scalable and modular production-grade NLP workflows.
- Integrated MLflow for experiment tracking and DVC for dataset/model versioning, ensuring reproducibility, traceability, and efficient model lifecycle management in production environments.

Full-Stack Wedding Management Platform

Tech Stack: Java, Spring Boot, React, JavaScript, RESTful APIs, MySQL, Apache Kafka, Docker, Git

- Built a full-stack wedding management platform using Spring Boot, React, Java, and MySQL, implementing scalable REST APIs and dynamic frontend components for seamless user experience.
- Developed core platform features including vendor listing, search, filtering, and review systems, enabling efficient vendor discovery and real-time user interaction.
- Worked in an agile, Git-based collaborative environment, contributing to frontend-backend integration, debugging, testing, and modular feature development across the application lifecycle.

Desktop Employee Management System

Tech Stack: C#, .NET Framework, Windows Forms, SQL Server

- Built a desktop-based Employee Management System using C# and .NET Framework with Windows Forms, enabling efficient employee record management and administrative operations.
- Implemented full CRUD functionality with SQL Server integration, supporting secure and reliable data operations including insert, update, delete, and retrieval.
- Designed a modular, event-driven OOP architecture with interactive UI components, ensuring structured backend logic and smooth user interaction in a business application.